

Warehouse Management System

Man-Days Estimation — Chiffrage

Effort estimate by functionality — code written by Claude Opus 4.8

Scope: New WMS module for a 3PL (ERP back-office + PWA scanner), integrated into the existing host

Basis: Bottom-up over 60 build items across 9 phases, calibrated to the agreed mockups

Code author: Claude Opus 4.8 — man-days are human-led effort (spec, review, integration, test)

Status: First-cut planning estimate, for client review

Date: 19 June 2026

1 · Headline Estimate

Estimated effort with code written by Claude Opus 4.8: **≈ 364 man-days** (planning / P50).

With a 25% contingency for the brownfield-integration unknowns: **≈ 456 man-days** (P80 — the figure to budget to).

For comparison, the same scope hand-coded by a traditional team is ≈ 970–1212 man-days. **Using Opus 4.8 removes roughly 62% of the effort** — the code is generated; the remaining man-days are human specification, review, host integration, testing and iteration.

All figures are human man-days (≈ one person-day each), not calendar time. Opus 4.8 runtime is a tool cost, not a man-day. See §2 for the method and assumptions.

Summary by phase

#	Phase	Items	Traditional md	With Opus 4.8 md
0	Foundation / Kernel	10	122	46
1	Master Data	8	59.5	19
2	Goods Reception	7	95	37
3	Putaway	3	43	17
4	Stock Visibility & Status	3	30	11
5	Stock-Out (Outbound)	8	116	46
6	Inventory Operations	8	108	42
7	Reports & Visibility	7	68.5	24
8	Hardening & Closure	6	66	24
	Engineering subtotal	60	708	266

Roll-up to the budget figure

Roll-up	Traditional md	With Opus 4.8 md
Engineering (sum of all build items)	708	266
+ Delivery overhead (PM, system test & UAT, iteration, release) — 37%	262	98
= Project subtotal (P50, planning)	970	364
+ Risk contingency (brownfield unknowns) — 25%	242	91
= Total to budget (P80)	1212	456
Optional — PWA offline sync (only if a v1 requirement, decision #5)	+ 48	+ 21

The optional offline-sync line is excluded from the totals above; add it only if offline scanning is confirmed as a v1 requirement (it is the single largest cost swing).

2 · Method & Assumptions

What a man-day means here

A man-day is one person-day (~8 h) of human engineering effort. Because Claude Opus 4.8 writes the code, that day is spent specifying the change, directing the model, reviewing and correcting what it generates, wiring it into the existing host, testing, and iterating with the client — not hand-typing the bulk of the code. The estimate assumes one engineer fluent in the host stack (.NET Core 3.1 / Angular 14 / SQL Server 2014) and skilled at driving Opus 4.8.

How each figure is built

Every build item is sized on the project estimation guide's complexity tiers (which sum backend, database, ERP UI, PWA UI and test effort). That yields a traditional human figure. An Opus-4.8 acceleration factor is then applied per tier — larger for boilerplate (CRUD), smaller for the host-integration and engine work where human review and debugging dominate:

Tier	Profile	Trad. md	× Opus	Opus md
S	CRUD master over the generic base service	4	0.30	1.0
S–M	small CRUD with a little logic	6.5	0.33	2.0
M	CRUD + non-trivial rules / one guard	9	0.35	3.0
M–L	multi-step, some cross-cutting calls	13	0.38	5.0
L	multi-step dual-channel flow, 3+ cross-cutting calls	17	0.40	7.0
XL	engine / orchestration, heavy edge cases	28	0.45	12.5

Tier baselines are calibrated against the project's estimation guide and the completed mockups as a reference class.

What is included

- Backend domain + services + APIs, database schema & migrations (SQL Server 2014), ERP screens (Angular/NgRx/Kendo), PWA scanner screens, and per-item tests.
- Integration into the existing host — Swagger client regeneration, gateway routing, DI wiring, identity/role reuse.
- Delivery overhead — project management, system & integration testing, client UAT support, environment/CI-CD, SQL migration runbook and release.
- A risk contingency keyed to the brownfield unknowns (SQL 2014 child-table modelling, the shared Auth module, .NET 3.1 / Angular 14 tooling, the open decisions).

What is excluded (price separately)

- Client portal and billing (out of v1 by decision).
- Label-printer / hardware integration and procurement (the mock prints client-side; production needs a print service).
- Infrastructure & hosting, data migration from any legacy WMS, end-user training & change management, and post-go-live support / SLA.

Key assumptions

- The functional scope and UX are locked — the Phase-2 mockups and the docs/ specs act as the specification, which removes most analysis/design effort (already reflected in the factors).

- Brownfield build: a new WMS module inside the existing ERP, reusing identity, data-access, gateway and UI kit.
- Online-only PWA for v1. Offline scanning (decision #5) is the separate optional line.
- Stay on the host runtimes (.NET Core 3.1 / Angular 14 / SQL Server 2014); EF migrations for schema; WMS-side tenant scoping.
- One delivery team. Figures are human man-days, not elapsed calendar time (parallelism shortens the calendar, not the man-days).

3 · Detail by Phase & Functionality

One line per build item. "Delivers" names the function or page(s) produced. Traditional = hand-coded human days; Opus 4.8 = human-led days with the model writing the code.

Phase 0 · Foundation / Kernel

One-time enabling platform — module, schema, and the shared services every process reuses.

Item	Delivers	Tier	Trad. md	Opus md
P00-S01	Backend module skeleton (7 projects) + DI / gateway route / Swagger client pipeline; Angular ERP + PWA project shells & routing	M	9	3
P00-S02	SQL Server 2014 schema foundation, EF DbContext, audit-ledger table, migrations + seed from the mock dataset	L	17	7
P00-S03	Tenant / site scoping service + WMS role/claim integration (touches the shared Auth module)	L	17	7
P00-S04	Stock-ledger service — append-only transaction log, written in-transaction by every action	M	9	3
P00-S05	Freeze service — the cross-screen frozen-scope guard	M	9	3
P00-S06	Capacity + segregation service — single source of truth for every bin write	M	9	3
P00-S07	Allocation engine core — FEFO / FIFO + reservation-aware availability	L	17	7
P00-S08	Reason-code service + approval / maker-checker framework	M	9	3
P00-S09	Work-item assignment facet (My-tasks / claim) + tracking-flag (lot/expiry/serial) plumbing	M	9	3
P00-S10	PWA shell — scan engine (camera / ZXing / manual / list), service worker, shared models & API client	L	17	7
Phase 0 subtotal			122	46

Phase 1 · Master Data

Mostly the CRUD recipe over the Phase-0 base classes — highly parallel.

Item	Delivers	Tier	Trad. md	Opus md
P01-S01	Clients (+ back-order policy), Suppliers, Carriers, Consignees	S–M	6.5	2
P01-S02	Sites (+ addressing levels, storage areas), Locations (+ capacity, area)	M	9	3
P01-S03	Products (+ tracking flags, category, weight, preferred storage), Categories, UoM	M	9	3
P01-S04	Packaging — shared templates + per-product packaging with per-level barcodes	M	9	3

Item	Delivers	Tier	Trad. md	Opus md
P01-S05	Reason Codes master (all domains)	S	4	1
P01-S06	Users + WMS roles/rights + client/site scope + client–user mapping	M	9	3
P01-S07	PWA lookups (location, product) — read-only	S	4	1
P01-S08	Bulk CSV import (products, locations)	M	9	3
	Phase 1 subtotal		59.5	19

Phase 2 · Goods Reception

First stock-creating process — ASN → receive → inspect → GRN, minting LPNs. Dual-channel.

Item	Delivers	Tier	Trad. md	Opus md
P02-S01	ASN management — CRUD, lines, derived open / partial / closed status	M	9	3
P02-S02	Receive against ASN + blind → mint LPNs, LPN split (per-each / per-pack), close-short, wrong-client guard	L	17	7
P02-S03	Mixed-pallet build (aggregate LPN + manifest label) + decomposition hand-off to Putaway	L	17	7
P02-S04	Inspection — quantity-based partial accept/reject, dispositions, reason codes	L	17	7
P02-S05	GRN — Goods Received Note (document / print, one per receipt)	M	9	3
P02-S06	Labels / reprint (LPN + pallet, recovery by ASN / recent, audited)	M	9	3
P02-S07	Reception edge cases — refuse-at-door, over-receipt approval, duplicate / damaged / expired guards, cold-chain	L	17	7
	Phase 2 subtotal		95	37

Phase 3 · Putaway

Directed slotting into bins, enforcing capacity / segregation / freeze. Dual-channel.

Item	Delivers	Tier	Trad. md	Opus md
P03-S01	Single-LPN directed putaway — ranked bins (home→consolidate→category→open), guards, partial split	L	17	7
P03-S02	Mixed-pallet decomposition — operator-chosen order → child LPNs	L	17	7
P03-S03	Damage-found reject, overflow-park, resume / partial-progress visibility	M	9	3

Item	Delivers	Tier	Trad. md	Opus md
	Phase 3 subtotal		43	17

Phase 4 · Stock Visibility & Status

The read model + status engine that Stock-Out and Reports both depend on.

Item	Delivers	Tier	Trad. md	Opus md
P04-S01	Stock-on-hand / availability read model + query API (the hot path for outbound & reports)	L	17	7
P04-S02	Stock Status screen + reason-coded status-change engine	M	9	3
P04-S03	Expose blocked / frozen / expiry / in-transit flags consistently across every stock payload	S	4	1
	Phase 4 subtotal		30	11

Phase 5 · Stock-Out (Outbound)

The most complex process — orders → allocation → pick/dispatch → delivery notes. Dual-channel.

Item	Delivers	Tier	Trad. md	Opus md
P05-S01	Outbound orders — CRUD, lines, ship-to consignee, full-stock-out, cancel order / line, status flow	M–L	13	5
P05-S02	Allocation surface — FEFO/FIFO, reservation-aware, short-pick, manual override	L	17	7
P05-S03	Pick / Dispatch (classic) — scan-confirm, serials-on-issue, carrier + proof-of-load, Delivery Note	L	17	7
P05-S04	Express Fulfil — one-pass allocate→pick→dispatch, role-gated	L	17	7
P05-S05	Back-order vs short-close + multi-shipment delivery notes	M	9	3
P05-S06	Ad-hoc / emergency dispatch — governance gate, role-gated + back-office approval queue	L	17	7
P05-S07	Return-to-vendor / client — non-consignee outbound	M	9	3
P05-S08	PWA pick / dispatch + ad-hoc + RTV parity	L	17	7
	Phase 5 subtotal		116	46

Phase 6 · Inventory Operations

In-warehouse stock operations incl. the freeze workflow; each card largely standalone.

Item	Delivers	Tier	Trad. md	Opus md
P06-S01	Move — intra-site relocation, partial split, guards	M–L	13	5

Item	Delivers	Tier	Trad. md	Opus md
P06-S02	Transfer — inter-site ship → in-transit → receive	L	17	7
P06-S03	Cycle Count — multi-location sheet → approve corrects stock; found / missing	L	17	7
P06-S04	Physical Inventory — freeze → count → post → unfreeze	L	17	7
P06-S05	Adjustments (qty) + Corrections (attributes) — approval-gated	M	9	3
P06-S06	Repack / split / merge / re-kit — genealogy + balance guard	L	17	7
P06-S07	Returns / put-back — dispositions (restock / via-putaway / quarantine)	M	9	3
P06-S08	Disposal / Scrap — approval-gated, terminal, maker-checker	M	9	3
	Phase 6 subtotal		108	42

Phase 7 · Reports & Visibility

Read-only over the stock read model + the ledger. ERP only.

Item	Delivers	Tier	Trad. md	Opus md
P07-S01	Stock on Hand	M	9	3
P07-S02	Transaction History / stock ledger	M	9	3
P07-S03	Expiry & Aging	S–M	6.5	2
P07-S04	Inbound & Outbound activity	M	9	3
P07-S05	Variance & Bin / Location Utilization	M	9	3
P07-S06	Lot / Serial Traceability + Stock Card / Product Ledger	L	17	7
P07-S07	Client Stock Statement	M	9	3
	Phase 7 subtotal		68.5	24

Phase 8 · Hardening & Closure

Production-grade non-functionals + carry-overs. (PWA offline sync is a separate optional line.)

Item	Delivers	Tier	Trad. md	Opus md
P08-S01	Export (CSV / Excel) across all reports	M	9	3

Item	Delivers	Tier	Trad. md	Opus md
P08-S02	Performance pass — indexes, query tuning, load tests on SQL 2014, grid paging	M–L	13	5
P08-S03	Security review — server-side scoping on every endpoint, role matrix, authz tests	L	17	7
P08-S04	Audit completeness — immutable ledger + before/after snapshots, reprint / override auditing	M	9	3
P08-S05	Operations dashboard wired to live rollups	M	9	3
P08-S07	Carry-over alignment sweep — shared capacity service, dispatch-txn parity, etc.	M	9	3
	Phase 8 subtotal		66	24

Optional — conditional on decision #5

Item	Delivers	Tier	Trad. md	Opus md
P08-S06	PWA offline sync — IndexedDB queue, sync-on-reconnect, conflict resolution (only if offline scanning is a v1 requirement — open decision #5)	XL	28	12.5

4 · Overhead, Risk & Adjusting the Estimate

Delivery overhead

Applied as a percentage of the engineering subtotal (it scales with the work):

Overhead component	%	Trad. md	Opus md
System test, integration & UAT	12%	85	31.9
Project management & coordination	12%	85	31.9
Client review & iteration loops	8%	56.6	21.3
Environment, CI/CD, SQL migration & release	5%	35.4	13.3
Total overhead	37%	262	98

Risk contingency

A flat 25% contingency takes the planning (P50) figure to a budget (P80) figure. The risk is concentrated in the brownfield integration — modelling the mock's nested data as SQL Server 2014 child tables, extending the shared Auth module for tenant/site scoping, the EOL .NET Core 3.1 / Angular 14 tooling, and resolving the five open integration decisions. A team may key the buffer per item instead (+15% known · +30% open question · +50% shared-module / unknown).

How to adjust

- The estimate is a model with four levers: the per-tier baseline days, the Opus-4.8 factor per tier, the overhead %, and the risk %. Change any one and the totals re-flow.
- The Opus-4.8 factors (0.30–0.45) are the central judgement. If early build items run faster or slower than assumed, recalibrate them after the first 2–3 items and the rest of the estimate follows.
- A line-by-line, adjustable spreadsheet version (one row per build item, with the levers on an assumptions sheet) is the natural next artifact if you want to tune the numbers interactively.

This is a first-cut, top-down-calibrated estimate for planning. It is not a fixed-price quotation; the figure to budget against is the P80 total in §1.